# SOC Network Analysis & Incident Report

ITSM TICKET EXPORT

| TICKET ID | SEVERITY / PRIORITY | ASSIGNEE | STATUS |
|---|---|---|---|
| INC-2026-1028-NET01 | P4 - Routine Diagnostic | Tier 1 SOC Analyst | Resolved / Closed |

## 1. EXECUTIVE SUMMARY

This document details the diagnostic network assessment, host enumeration, name resolution testing, and live packet capture analysis executed on the target host interface. The investigation validated host baseline connectivity, local gateway routing, external domain resolution via DNS, and service exposure on remote scan targets. All diagnostic controls passed standard operational thresholds without detecting malicious activity or unauthorized protocol behaviors.

## 2. HOST IDENTIFICATION & INTERFACE CONFIGURATION

| Parameter | Observed Value | Verification Method |
|---|---|---|
| **Active Interface** | `eth0` (Loopback `lo` active) | System Network Stack Inspection |
| **IPv4 Address** | `10.0.2.15` | Command: `ip a` |
| **Default Gateway** | `10.0.2.2` | Command: `ip route` |
| **Privilege Requirement** | Elevated ( `sudo` required for raw sockets) | OS Security Policy Verification |

## 3. CONNECTIVITY & DOMAIN RESOLUTION TRIAGE

Network path verification was executed against public endpoints to confirm outbound reachability, transit latency, and name resolution integrity:

- **ICMP Reachability ( `ping` ):** Successful delivery to target `google.com` with 0.0% packet loss. Round-trip times recorded: Min = 65.405 ms, Max = 119.728 ms, Average = **93.507 ms**.
- **DNS Lookup ( `nslookup` ):** Queries successfully resolved `google.com` to IPv4 target `172.217.26.78` and global IPv6 endpoint `2404:6800:4017:80b::200e`.
- **Route Tracing ( `traceroute` ):** Confirmed layer-3 transit egressing via local default gateway ( `10.0.2.2` ) reaching remote infrastructure within expected minimal hop counts.

> **SOC TRIAGE SUMMARY**
> All network interface bindings and layer-3 transit controls are operating within baseline specifications. No packet drop or DNS spoofing indicators were observed.

## 4. RECONNAISSANCE & PORT ENUMERATION (NMAP)

Reconnaissance scans targeting standard security testing host `scanme.nmap.org` were conducted to evaluate port visibility and service finger-printing features ( `-sV` ):

| Target Host | Detected Port | Service Protocol | Operational Status |
|---|---|---|---|
| `scanme.nmap.org` | Port 22 | SSH | Open / Listening |
| `scanme.nmap.org` | Port 80 | HTTP | Open / Listening |

## 5. PACKET CAPTURE & TELEMETRY ANALYSIS (TCPDUMP)

Live packet captures on interface `eth0` were performed using `sudo tcpdump -i eth0` to inspect link-layer framing and active IP communications:

- **Link Encapsulation:** Captured frames confirmed standard Ethernet ( `EN10MB` ) formatting across active network sockets.
- **Traffic Distribution:** Monitored streams contained both local subnet traffic and outbound public transit endpoints.
- **IPv6 Telemetry (Bonus Capture):** Identified active IPv6 address allocations, including Unique Local Addresses ( `fd17:...` ), Link-Local Addresses ( `fe80:...` ), and remote Google IPv6 destinations.

## 6. TECHNICAL KNOWLEDGE BASE & GOVERNANCE

**Q1: Why do specific diagnostic network commands require elevated privileges (sudo)?**

Commands such as `tcpdump` or low-level `nmap` SYN scans require root privileges because they interact directly with raw network sockets and network interfaces in promiscuous mode. Restricting raw socket creation to superusers prevents unprivileged applications from sniffing unencrypted traffic or injecting arbitrary network frames.

**Q2: What is the significance of Nmap in Security Operations?**

Nmap (Network Mapper) is a foundational tool for attack surface management, asset discovery, and vulnerability assessment. SOC analysts utilize Nmap to identify open ports, verify running service versions, audit firewall rules, and detect unauthorized services bound to network interfaces.

**Q3: How does Tcpdump support active security monitoring and threat investigation?**

Tcpdump provides real-time packet-level visibility into traffic passing through an interface. In incident response, analysts use Tcpdump to capture raw PCAPs, inspect suspicious TCP flags, analyze malware command-and-control (C2) payloads, and reconstruct network events during forensic investigations.